Session Objectives

1. Explain how epidemiology, microbial virulence factors, and host-defense mechanisms contribute to the prevention and development of zoonotic infections affecting the skin, blood, lymphatic systems, and other sites.
2. Differentiate among key zoonotic pathogens based on their unique microbiologic characteristics and their involvement in infections of the skin, blood, lymphatic systems, and other anatomical sites.
3. Apply common clinical microbiology testing procedures to identify major pathogens responsible for zoonotic infections of the skin, blood, and lymphatic systems.
4. Interpret microbiological results to support accurate clinical diagnosis of zoonotic diseases.

Source: Course Syllabus

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Pathogens/Microorganisms Covered

1. *Yersinia*
2. *Bacillus anthracis*
3. *Borrelia*
4. *Leptospira*
5. *Rickettsia*
6. *Anaplasma* and *Ehrlichia*
7. *Coxiella*
8. *Francisella*
9. *Brucella*
10. *Other: Zoonotic influenza, Salmonellosis, West Nile, SARS, Lyme Disease*

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ZOONOTIC INFECTIONS

1. Infection acquired from any type of animal:
   a. Vertebrates (mammals; fishes, etc.)
   b. Invertebrates (insects / arthropods)

2. Often acquired from pets

3. Children are usually the most affected group

4. Occupational exposure (e.g. veterinarians; landscapers, national parks workers, and people who work in meat-packing plants)

Common Routes of exposure

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2. Inhalation (from aerosolized microbes)
   - Anthrax: from spores
   - Q fever [mostly from farm animals]
3. Ingestion
   - Cryptosporidiosis [contaminated water]
   - Leptospirosis [contaminated water / food]
   - Anthrax [contaminated food]

Yersinia – Clinical Diseases

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- *Y. pestis:*
  - Bubonic: Infected flea bite → fever and bubo (axillary/inguinal lymphatic swelling) 7 days later → 75% death rate w/o treatment
  - Pneumonic: 2-3 day incubation, fever and pulmonary symptoms w/n 1 day; 90% death rate w/o treatment
- *Y. enterocolitica*: Enterocolitis + Pseudoappendicitis (kids)
  - Avg. incubation = 4-6 days → bloody diarrhea, fever, abdominal pain x 1-2 weeks
  - Terminal ileum, mesenteric lymph nodes
  - Other: septicemia, intra-abdominal abscess, arthritis, hepatitis, osteomyelitis

Bacillus anthracis

- Gram + rod, obligate spore-forming aerobe, encapsulated, lethal cause of anthrax; bioterrorism
- Physiology/Structure:
  - Long chains of single/double rods
  - Three toxins on one plasmid
    - Protective Antigen (PA)
    - Edema Factor (EF): CaM-dependent AC → ↑cAMP → edema
    - Lethal Factor (LF): Zn metalloprotease
- LF → macrophage apoptosis
- LF + EF → inhibition of innate immunity
- Capsule: poly D-glutamic acid

PA + EF = Edema Toxin

B. anthracis: Epidemiology

- Unvaccinated herbivores (livestock) → accidental human infection
- 2001: USPS purposeful contamination (spore-induced infection of a postal employee)
- Route: inoculation/ingestion/inhalation
- Inhalation = “wool-sorter’s disease”
- Spores are highly resilient
- No person-person transmission

B. anthracis – Clinical Diseases

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- Cutaneous Anthrax
  - Painless papule → ulcer w/surrounding vesicles → necrotic eschar (+ painful edema and lymphadenopathy [LAD])
  - *Necrosis is due to Lethal Factor (LF)*
- GI Anthrax:
  - Upper: regional LAD, sepsis, edema
  - Lower: Nausea, vomiting, malaise
- Inhalation Anthrax: alveolar macrophages deliver spores to mediastinal LN's (latency period) → nonspecific URI → mediastinal LAD, fever, massive edema → shock + death within 3 days

"Cutaneous Anthrax" N Engl J Med 2009; 361:178
DOI: 10.1056/NEJMicm0802093

B. anthracis – Lab + Clinical Diagnosis

- Visible on Gram stain: long, thin rods, Gram+, chains
- Low-CO2 culture can grow spores
- Difficult distinction from *B. cereus* (*B. anthracis* is non-hemolytic, nonmotile)
- India ink exposes capsule
- "medusa head" colonies on sheep blood agar

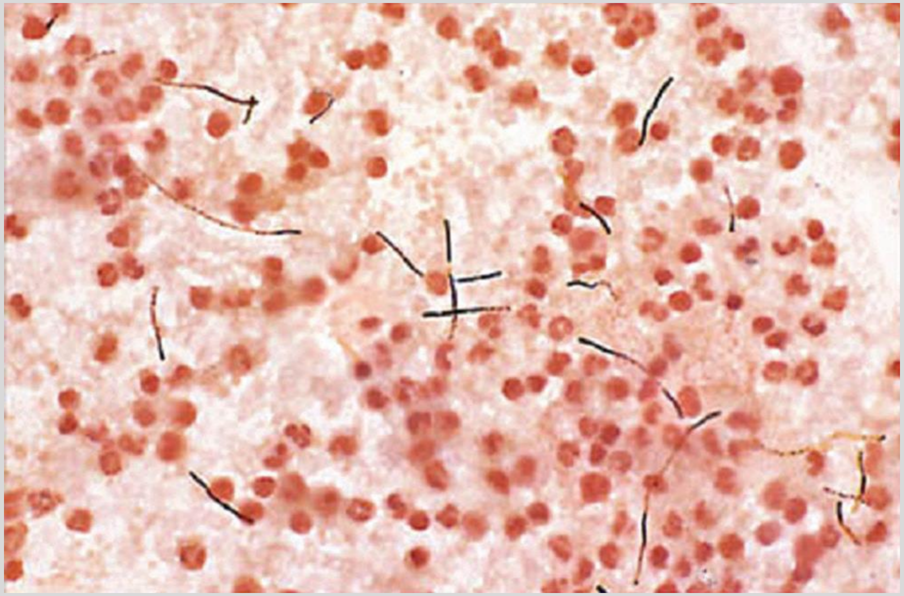

Fig. 20-2 (Murray): *B. anthracis* in blood from patient with pulmonary anthrax

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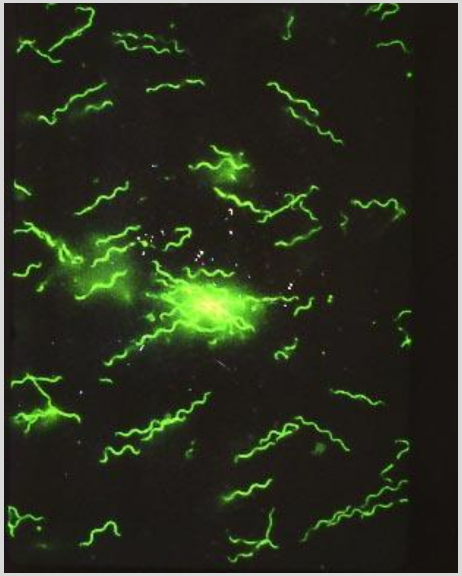
B. anthracis – Treatment, Prevention, Control

- Empiric treatment = Ciprofloxacin or Doxycycline + 1-2 additional (Rifampin, Vancomycin, Penicillin, Imipenem, Clindamycin, Clarithromycin)
- Amoxicillin for cutaneous anthrax
- Vaccination: animals themselves, residents of endemic regions, handlers of animal products from endemic regions, and certain military personnel.

Borrelia

- Gram –ve spirochete (microscopy)
- *B. burgdorferi* & *B. recurrentis*
- Transmitted by *Ixodes* ticks
- Diseases: (1) Lyme disease (caused by *B. burgdorferi*) & (2) relapsing fever, due to *B. recurrentis*
- Poor Gram staining → stain with Giemsa or Wright



B. burgdorferi on fluorescence microscopy

Lyme Disease: Clinical Manifestations

- Erythema migrans rash ~5 cm diameter <ins>*or*</ins> ≥1 late manifestation (MSK, nervous system, cardiovascular) + lab confirmation
- Lab definition:
  - Isolation of *B. burgdorferi* [not usually done]
  - Dx levels of IgM or IgG Ab's
  - Significant increase in Ab titer between acute/convalescent stages
- Treatment: doxycycline

Leptospira – Pathogenesis & Immunity

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- Subclinical infection
- Mild flu-like illness
- Systemic disease (Weil Disease)
- Penetration of mucous membranes, cuts
- Location:
  - Early: blood, CSF
  - Late: urine
- Some clinical manifestations from immunologic reactions

Leptospira: Clinical Disease

- Infection through skin or conjunctiva
- 1-2 week incubation → flu-like illness (fever, myalgia, chills, HA, vomiting, diarrhea) →
  - remission *or*
  - more severe illness: HA, myalgia, chills, abdominal pain, conjunctival suffusion →
  - Severe disease = vascular collapse, TCP, hemorrhage, hepatic/renal dysfunction
- Weil Disease = severe, icteric disease
  - No hepatic necrosis, usually full renal recovery

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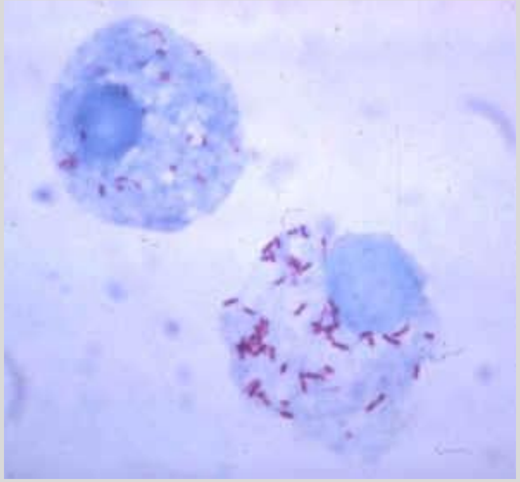
Rickettsia rickettsii

- Rocky Mountain Spotted Fever
- Obligate intracellular Gram - rod (doesn’t stain well)
- Stains best with Giemsa or Gimenez stain
- Arthropod vectors, rodent reservoirs
  - *Dermacentor* (dog tick)
- Transovarian transmission permits ticks to be hosts and vectors



Rockettsia growing inside cultured cells: appear as short
rods using a special stain

Rickettsia rickettsii

- Endothelial adherence via OmpA (Virulence Factor)
- Vascular leakage → hypovolemia, hypoproteinemia
- Cytokine-mediated killing via CD8+ T-cells
- Tick bite → macular rash (wrists, arms, ankles → trunk) 3 days post-tick bite
  - 7 days post-tick bite: high fever, HA, malaise, myalgia, N/V/Abdominal Pain, diarrhea
- Sequelae:
  - rash turns petechial (poor prognosis)
  - Renal and pulmonary failure, neurologic complications, cardiac abnormalities
  - 10-25% fatality rate if untreated
  - Endotoxin related

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Rash of RMSF in early (top) and late (bottom) stages. Image: CDC

Anaplasma and *Ehrlichia*

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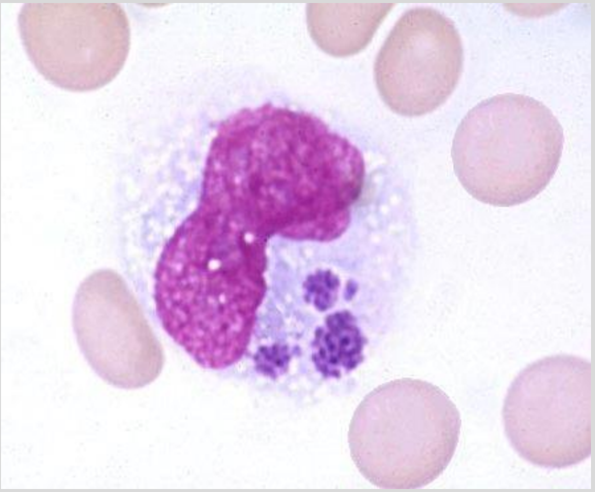
- Intracellular: mostly granulocytes (PMNs-*Anaplasma*), or monocytes – *Ehrlichia chafeensis*)
- Prevent phagolysosome fusion
- Gram –ve but no PG/LPS
- Giemsa, Wright staining
- Cross reaction common on serology
- Tick transmission, vertebrate hosts

Genus	**Vector**	**Host**
Erhlichia	Lone Star Tick (*A. americanum)*	White-tailed deer
Anaplasma	*Ixodes*	Small rodents

Human Monocytic Ehrlichiosis

- *Ehrlichia chafeensis*
- Emerging disease (1986) in US
- Tick bite → 1-2 weeks later: high fever, HA, malaise, myalgia → late-onset rash
  - Most patients develop leukopenia, thrombocytopenia, elevated transaminases
  - Immunocompromised patients can develop sepsis
- Most patients require hospitalization (2-3% mortality)
- Reservoir white-tailed deer



Infected monocyte with *3 morulae*

Anaplasmosis

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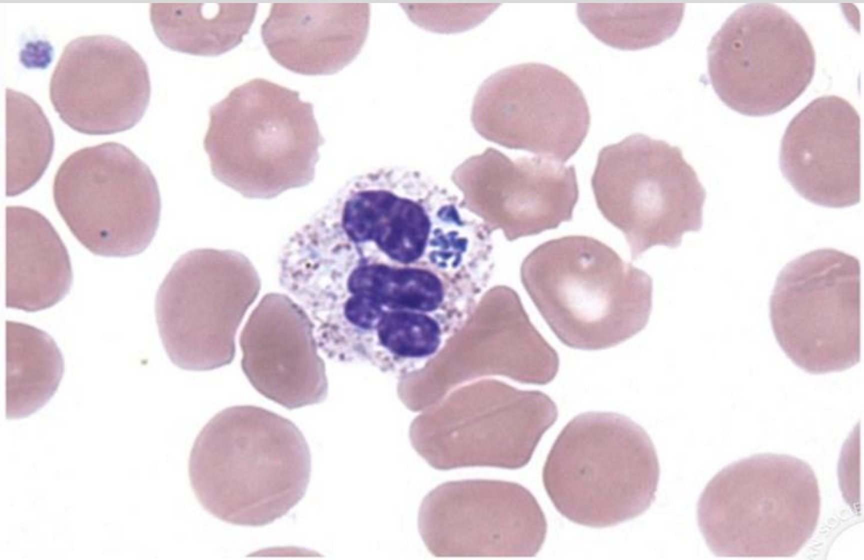
- *Anaplasma phagocytophilum*
- Mostly granulocytes (PMNs)
- Tick bite → 5-10 days later: high fever, HA, malaise, myalgia
  - TCP, leukopenia, ↑transaminases (same as Ehrlichiosis)
- Sequelae: peripheral neuropathy

A neutrophil infected with *Anaplasma phagocytophilum*

Blood 2012; 120:4911-4911



Pozdnyakova O , Dorfman D M Blood 2012;120:4911-4911

Ehrlichiosis + Anaplasmosis: Diagnosis & Treatment

- Giemsa stained preps only useful in first week of infection; Gram stain ineffective
- Nucleic acid amplification and serology
  - Titers increase 3-6 weeks after presentation
  - Serology cross-reaction: RMSF, Q fever, Lyme, Brucellosis, EBV
- Treatment: doxycycline

Coxiella

- Mild symptoms → abrupt high-grade fever, fatigue, HA, myalgia
  - Severe sequelae (~5%): hepatitis, pneumonia, fevers
  - Look for granulomas in involved organs
  - Chronic: usually subacute endocarditis if immunosuppressed/preexisting valvulopathy

Coxiella: Diagnosis, Treatment

- Most common: serology (immunofluorescence antibody)
- Treatment: doxycycline for 14 days
  - Chronic: doxy. + hydroxychloroquine
- Vaccine: inactivated, whole-cell organisms
  - Uninfected animals, animal workers; no booster

Francisella

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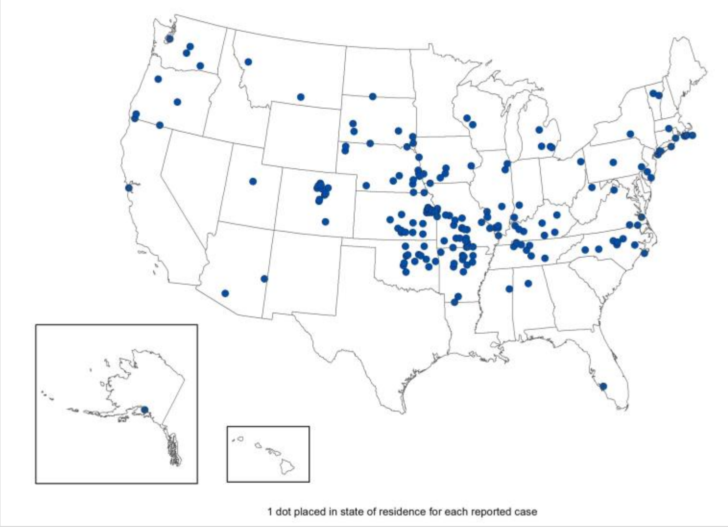
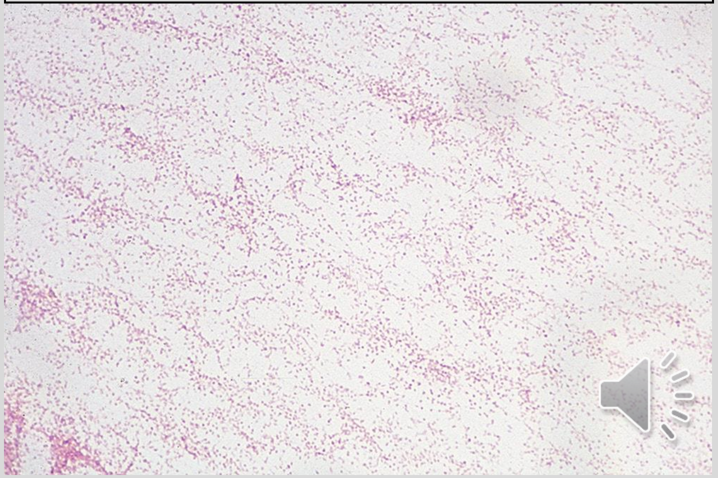
- Multiple species in genus, most important is *F. tularensis*
- Aerobic Gram -ve, faintly staining coccobacillus, intracellular
- Culture: fastidious (cysteine; >3 days for growth)
- Culture hazardous
- Capsule
- Transmitted by ticks & deer flies; exposure to rabbits



Above: 2023 Infections (CDC)
Below: *F. tularensis* Gram stain (Murray Fig. 29-4)



Tularemia

- AKA Glandular Fever/Rabbit Fever/Tick Fever/Deerfly Fever
- Type A infections: rabbits, hares, ticks (*Ixodes, Dermacentor, Amblyomma*)
- Most common manifestation: ulceroglandular (papule → necrotic ulcer with lymphadenopathy and bacteremia)
- Other types: oculoglandular, pneumonic, typhoidal, oropharyngeal, gastrointestinal
- In US: primarily in OK, AR and MO

Glandular Tularemia (NEJM)

F. tularensis: Lab Diagnosis + Treatment

- Extreme biohazard in specimen collection
- Growth on buffered charcoal yeast extract (BCYE) agar w/cysteine supplement
  - BCYE also used for *Legionella*
- Ab titers usually used for Dx
- Treatment of choice: Gentamicin
  - Alternate: Doxycycline, Ciprofloxacin
- Prompt tick removal can prevent infection

Brucella Lab Diagnosis

- ≥4-fold titer increase from acute to convalescent phase of illness (≥2 weeks apart)
- Culture on blood agar or other enrichment media x 2 weeks before determining negativity
- Presumptive Dx: 4-fold increase *or* any single titer higher than 1:160

Brucella Clinical Syndromes

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- Localization to organs with ↑erythritol
  - Breast, uterus, placenta, epididymis
- Symptoms 1-3 weeks after exposure: malaise, chills, sweat, fatigue, weight loss, nonproductive cough
- Fever can wax and wane: undulant fever
- Inadequate Tx produces chronic infection 3-6 mos. after antibiotics stopped:
  - Bone (osteomyelitis), spleen, liver are other sites of infection

Brucella Treatment

- Doxycycline + Rifampin
- Pregnant women and children ≤8 years old: trimethoprim-sulfamethoxazole instead of doxy.
- Treat for ≥6 weeks
- Prevention: animal vaccination (live attenuated organisms), avoidance of unpasteurized dairy products

Case 1

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A 35-year-old male presents to his family physician with a 2 week history of joint pain in his left knee area. He recalls having a low-grade fever and malaise 4 weeks earlier. He works as a landscaper on Eastern Long Island. Upon questioning, he reveals that he had a rash about a month earlier, and he shows the physician a photo of the rash (shown below):

-What is the likely pathogen and pathogenesis of this disease?

-How does this disease evolve clinically?

Summary slide

- A wide variety of zoonotic infections are caused by blood-borne pathogens.
- Many of these agents are transmitted by various insect or arthropod vectors, or certain animal reservoirs.
- Transient skin lesions are often associated with these infections with extracutaneous disease occurring via hematogenous spread of the microbial agent.

Source: Course Syllabus